and SIGINT to produce geo-registered tracks and classification confidence levels.

- *Motion planning and control*: Real-time path planning (including 4D trajectory optimization), collision avoidance (sense-and-avoid), and guidance laws that respect dynamic constraints and no-fly/keep-out zones.
- *Distributed coordination middleware*: Real-time publish/subscribe or DDS-style middleware (or ROS2 variants) enabling low-latency state sharing, consensus algorithms for swarm decisions, and distributed tasking.
- *Cybersecurity and resilience layer*: Secure boot, attestation, anomaly detection (behavioral and ML-based), encrypted comms, anti-spoofing, and intrusion response that can shift to autonomous safe modes.
- *FDIR and prognostics*: Continuous monitoring and automated recovery controllers that handle sensor failures, actuator degradation, and partial system loss with logged audit trails.
- *Human–Machine Interface (HMI)/GCS software*: Intent-based mission planning tools, visualization, explainable AI summaries, and supervisory controls to allow human override and mission parameter adjustments.
- *Simulation and verification tools*: Onboard/ground digital twins and physics-based simulators for plan validation and safe mission rehearsal.

### 5.3.4 System integration and functional operation

Integration converts components into a dependable system through engineering practices, verification, and operational concepts: